AI Engineering: Fast Track Roadmap for SWEs 2026

The following is the roadmap for software engineers who want to transition into an **AI Engineer** role quickly in 2 months. It includes FREE learning resources for technical skills (or tool skills) and soft (or core) skills 🏄

This roadmap is designed based on the analysis of hundreds of AI jobs and our own experience of working on AI projects at Atliq Technologies (https://www.atliq.com/) where we have worked on 25+ AI projects in last two year alone. More than 90% of our clients at Atliq are SMEs (Small to medium size enterprises) based in USA.

Week 1: AI & Python Foundations

- **Topics**
  - AI Basics
    - Understand AI landscape by developing basic understanding of ML, DL, NLP, Gen AI and Agentic AI
  - Python is THE programming language for AI. Briefly learn the following topics,
    - Python Datatypes, Functions, Exceptions
    - Loops, Control Flow, File Handling
    - OOPs in Python
    - REST & FastAPI Basics: Calling APIs, JSON Responses, Decorators
    - Explore groq's python sdk
  - For your understanding, here we've included a code snippet in Python that calls LLM Api,

```python
from openai import OpenAI

client = OpenAI(api_key="YOUR_API_KEY")

response = client.chat.completions.create(
    model="gpt-4o-mini",
    messages=[
        {"role": "user", "content": "Explain AI in one sentence"}
    ]
)

print(response.choices[0].message.content)
```

- AI First Engineering Fundamentals with **Cursor, Claude or Antigravity**.
- Gen AI Basics:
  - Gen AI vs Agentic AI: What's the difference?
  - How do LLMs work?
  - LLM Configurations:
    - Context Window
    - Temperature
    - Top-p
    - Top-k
  - Embeddings & Vector Database
  - Use cases of Vector Database
  - Get some hands-on experience with **ChromaDB, Qdrant, Pinecone or Weaviate.**

- **Learning Resources**
  - Python:
    - AI Basics: https://www.youtube.com/watch?v=VGFpV3Qj4as
    - Python Tutorials (Codebasics) on YouTube (first 16 videos) - https://bit.ly/3X6CCC7
    - Corey's Python Tutorials: https://bit.ly/3uqUgaZ
    - AI First Development (Choose One):
      - Cursor: https://youtu.be/5tvsM6l7TiU?si=jRje7FfoLtXX9Z9O
      - Antigravity: https://youtu.be/1Q5sWISByuw?si=6rzZcfAyqCQUZuEm
      - Claude Code: https://youtu.be/HpnISrNvo0U
    - Groq API: https://youtu.be/S53BanCP14c?si=Uh9ukLFm9gKruDOb
  - GenAI:
    - LLMs Explained briefly: https://youtu.be/LPZh9BOjkQs?si=UzuwALmsvVrZo0q5
    - Vector DB Simple Explanation: https://youtu.be/72XgD322wZ8?si=KLE4W8QvFuhGzHEn
    - ChromaDB in Python Basics: Chroma Python Basics
  - **EXTREMELY IMPORTANT**: Use Claude Code 😎 as your personal tutor in case you have questions or facing issues

- **LinkedIn - Core Skill**

☐ Create a professional-looking LinkedIn profile.

- Have a clear profile picture and banner image.

- Add tags such as: Open to work etc.

☐ Use this LinkedIn Checklist to create a profile: Click here.

- **Assignment**

☐ Code a simple FastAPI Server to display a motivational Quote

☐ Build a simple search engine using Chroma or Qdrant

☐ Create a professional-looking LinkedIn and X profile.

- **Why LinkedIn and X is important?**
  - Imagine you are skilled at making samosas. You open a shop in a village where the footfall is very low. You are now struggling to sustain the business. Your best friend then gives you advice to move the shop to a busy street in a big city. You move it to a city and now your business is flourishing.
  - LinkedIn is like that busy street where your profile will be noticed by hundreds of people. Creating good looking profile (i.e. opening a shop) is only a first step, you need to build strong online credibility (i.e. making tasty samosas and providing good customer service) which we will discuss in the further sections.
  - Many people have this wrong mindset that "I will learn technical skills first and work on LinkedIn at last". Building online credibility will take a long time hence it is better to start working on it from day 1 👍



Week 2: RAG Fundamentals & LangChain

- **Topics**
  - What is RAG?
  - How to build a RAG Pipeline
  - Document Parsing & Chunking using Docling
  - LangChain Basics
  - ChatModels in LangChain
  - LangChain Integrations
  - RAG using LangChain
  - Tool Binding
- **Learning Resources**
  - GenAI Crash Course: https://bit.ly/3Fn7Zoh
  - Q&A System using LangChain: https://youtu.be/AjQPRomyd-k?si=z4vZchMM-rjLyZ9H
  - Rocky Chatbot (langchain + openAI): https://www.youtube.com/watch?v=MoqgmWV1fm8

- RAG + LangChain by pixegami: RAG + Langchain Python Project: Easy AI/Chat For Your Docs
- Docling Tutorial by RedHat: How Docling turns documents into usable AI data

- **Soft Skills – Online Credibility through X**
  - Andrej Karpathy: https://x.com/karpathy
  - Boris Cherny: https://x.com/bcherny
  - Yann Lecun: https://x.com/ylecun
  - OpenAI: https://x.com/OpenAI
  - Increase engagement.
    - Start commenting meaningfully on AI and career-related posts.
    - Helps network with others working in the industry build connections.
    - Learning and brainstorming opportunity.

  - Remember ***online presence is a new form of resume***

- **Fasttrack Your Learning**
  - AI Engineering Bootcamp for Software Engineers (Live): https://codebasics.io/bootcamps/ai-engineering-bootcamp-software-engineers

- **Business Fundamentals - Soft Skill**
  - Learn business concepts from ThinkSchool and other YT Case Studies
  - Example: How Amul beat competition: https://youtu.be/nnwqtZiYMxQ

- **Assignment**

☐ Code a RAG pipeline in pure python (can use docling for document parsing)
☐ Code a simple LangChain RAG Chatbot for Healthcare Sector
☐ Write meaningful comments on at least **10 AI related LinkedIn or X posts**
☐ Note down your key learnings from **3 case studies** on ThinkSchool and share them with your friend.

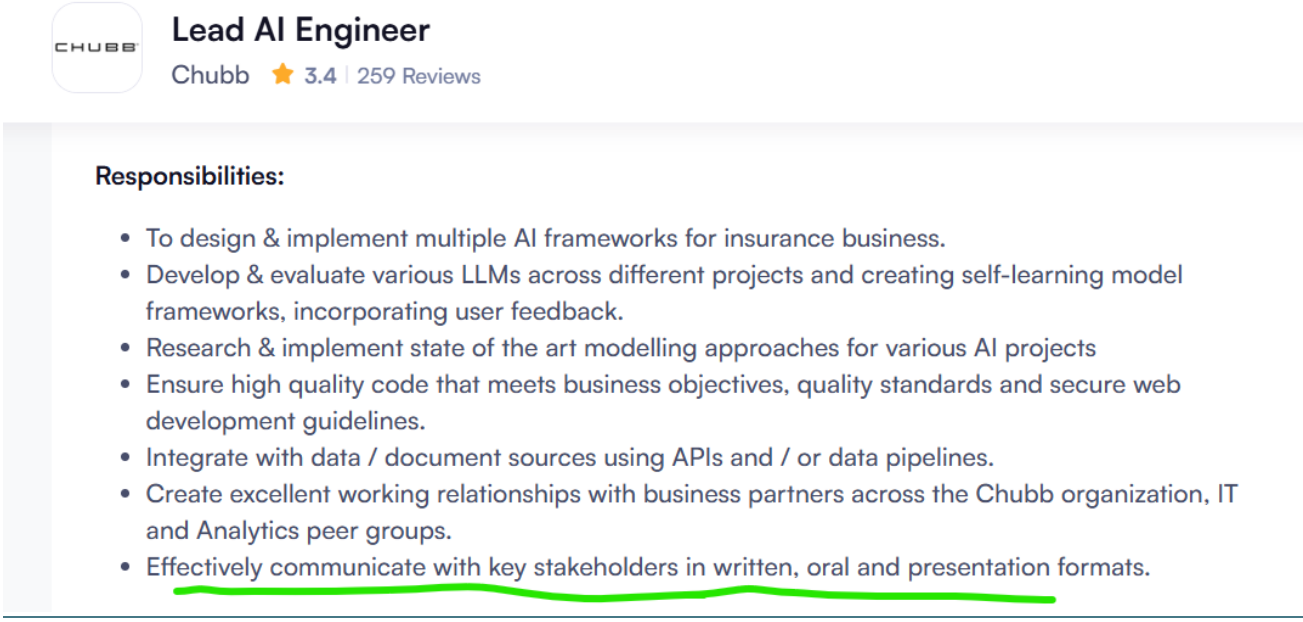
Week 3: Agentic AI Fundamentals

- **Topics**
  - What is an AI Agent?
  - ReAct: Reasoning + Action Loop
  - Building AI Agent in LangChain
  - Routing in AI Agents
  - AI Agent Security
  - Gaurdrails & Evaluation
- **Learning Resources**
  - What is AI Agent & how does it work: https://www.youtube.com/watch?v=15_pppse4fY
  - Building AI Agent in Agno: https://www.youtube.com/watch?v=EUey9L9sgzE
  - LangChain Docs: Agents - Docs by LangChain
  - Guardrails in Agents: https://youtu.be/rMUycP_cp9g?si=2GsIPWWwrrPxiZ3F
  - Evaluations with RAGAS; Evaluate AI Agents in Python with Ragas

- **Core/Soft Skills**
  - Presentation skills
    - Death by PowerPoint: https://youtu.be/Iwpi1Lm6dFo

- **Assignment**

  ☐ Build an AI Agent in Langchain, add guardrails & evaluate using Ragas
  ☐ Write **2** meaningful blog posts on the AI tech topic. For example, how CNNs work? Writing blogs or creating videos will help you build "*Teaching skills*" which will be helpful when you are talking to a non-tech business stakeholder while working as an AI engineer. Check the job post where they specifically look for strong stakeholder communication skills:

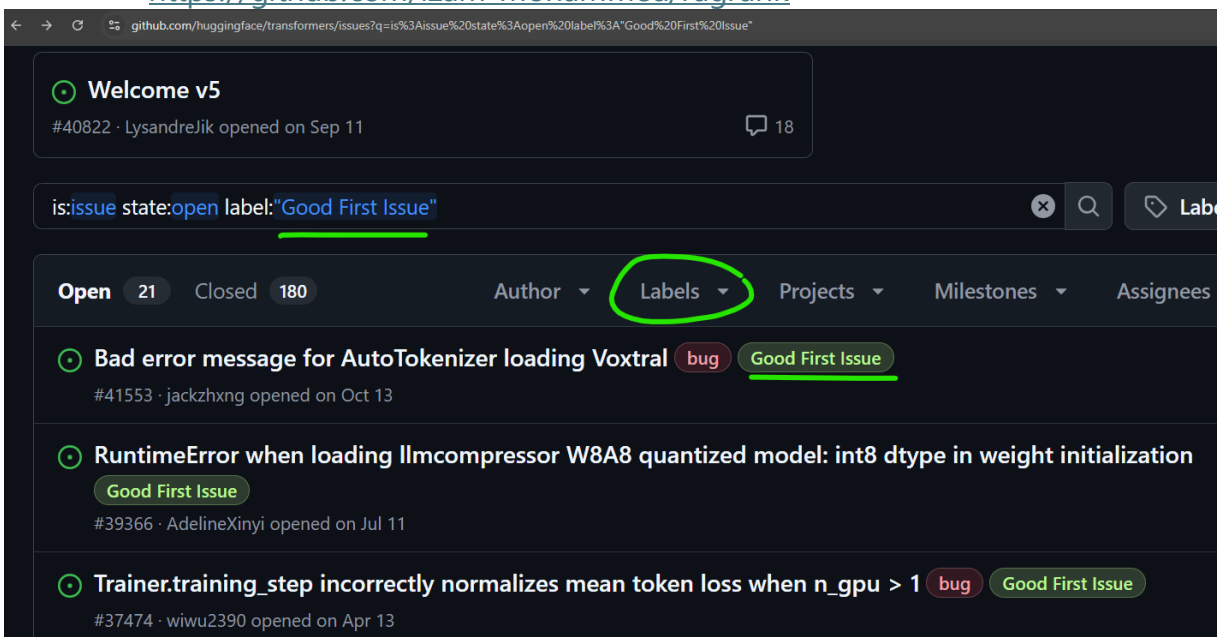


Week 4: AI Application Observability & Deployment

- **Topics**
  - Introduction to LangSmith
    - Traces
    - Runs
    - Threads
    - Cost Tracking
  - Agentic AI deployment
  - AWS AgentCore
  - Azure AI Foundry
- **Learning Resources**
  - LangSmith tutorial by langchain: https://youtube.com/playlist?list=PLfaIDFEXuae2CjNiTeqXG5r8n9rld9qQu&si=kwcs_yLt5Xerrd9p
  - AWS AgentCore tutorial: https://www.youtube.com/watch?v=cTBGIKAckKE
  - Microsoft AI Studio: https://youtu.be/fQ9RFR1KTbY?si=V0YFox990-dX41pf

- **Start contributing to opensource**
  - Use ChatGPT, Claude Code as your friend to get help on opensource contribution

- Find any repository, check issues and filter for Good first issue. Here is the screenshot from huggingface Github repo
- You can also try some simple repositories such as,
  - https://github.com/Mindinventory/MindSQL/issues
  - https://github.com/izam-mohammed/ragrank



- **Assignment**
  ☐ Deploy the Agent built in previous week on Azure or AWS

Week 5: Multi-Agent Systems & Context Engineering

- **Topics**
  - LangGraph
    - Introduction to LangGraph
    - State, Nodes & Graph in LangGraph
    - Orchestrating AI Workflow in LangGraph
    - Routing with Conditional Graph
    - MultiAgent System in LangGraph
    - Memory & human in the loop
  - Context Engineering
    - Model Context Protocol
    - How to build MCP Server & Client
    - What is context engineering/
    - How to implement it in Langgraph?

- **Learning Resources**
  - LangGraph Crash Course: https://youtu.be/CnXdddeZ4tQ?si=Ad2ypktxwrGJNtBY
  - Agentic AI project in LangGraph: https://www.youtube.com/watch?v=SP-b_G74Nuk
  - Human in the Loop: https://docs.langchain.com/oss/python/langchain/human-in-the-loop
  - What is MCP: https://youtu.be/tzrwxLNHtRY
  - Build your MCP server: https://youtu.be/jLM6n4mdRuA
  - Context Engineering: https://youtu.be/lcpWbg0jiYU?si=-Zzc7imlbp77FCWo

- **Assignment**
  ☐ Build a multi-agent system to automate marketing tasks
  ☐ Build an MCP Server for a public API

Week 6: Advanced AI Engineering

- **Topics**
  - Optimizing Cost of AI Apps
    - Response Caching
    - Rate limiting
    - Model Selection & Cascading
  - MultiModal RAG
  - Finetuning LLMs
    - What is LoRa & QLoRa?
    - When to finetune your model?
    - How to Finetune with Unsloth
  - How to run models locally with Ollama
- **Learning Resources**
  - Caching: Optimize RAG Resource Use With Semantic Cache - YouTube
  - Building Effective Agents: Building Effective AI Agents \ Anthropic
  - Finetuning Course: https://www.youtube.com/watch?v=IlvORO248Zs
  - Run deepseek r1 locally with Ollama: Run Deepseek R1 Locally

Week 7: Declarative AI using DSPy

- **Topics**
  - DSPy: Signatures, Modules Adapters

- Optimizers
- RAG in DSPy
- Building an Agent with DSPy

- **Learning Resources**
  - DSPy by AI Engineer: https://www.youtube.com/watch?v=-cKUW6n8hBU
  - Official Docs: Tutorials Overview - DSPy

- **Assignment**

  ☐ Build the lovable clone project using DSPy

Week 8: Azure or AWS 🌥

Cloud skills (especially Azure) are essential especially for experienced AI engineer roles. When we analyzed AI engineer jobs, we found Azure to be at the top among all cloud providers when it comes to AI.

- **Topics**
  - Azure fundamentals: Resource groups, subscriptions, regions, storage, IAM
  - Compute and deployment services: azure functions, azure app service, azure Kubernetes service (AKS)
  - AI, ML: Azure ML, OpenAI service, Azure AI Foundry
  - MLOps and Monitoring: Azure ML Pipelines, model monitoring

- **Learning Resources**
  - There are tons of free resources you can find on the internet. We do not have specific suggestions at this point but some day in future we plan to produce free videos on these topics on our YouTube channel!

- **Fasttrack Your Learning**
  - AI Engineering Bootcamp for Software Engineers (Live): https://codebasics.io/bootcamps/ai-engineering-bootcamp-software-engineers

Tips for effective learning

- **Spend less time in consuming information, more time in**
  - Digesting
  - Implementing
  - Sharing

- **Group learning**
  - Use **partner-and-group-finder** channel on codebasics discord server for group study and hold each other accountable for the progress of your study plan. Here is the discord server link: https://discord.gg/r42Kbuk